

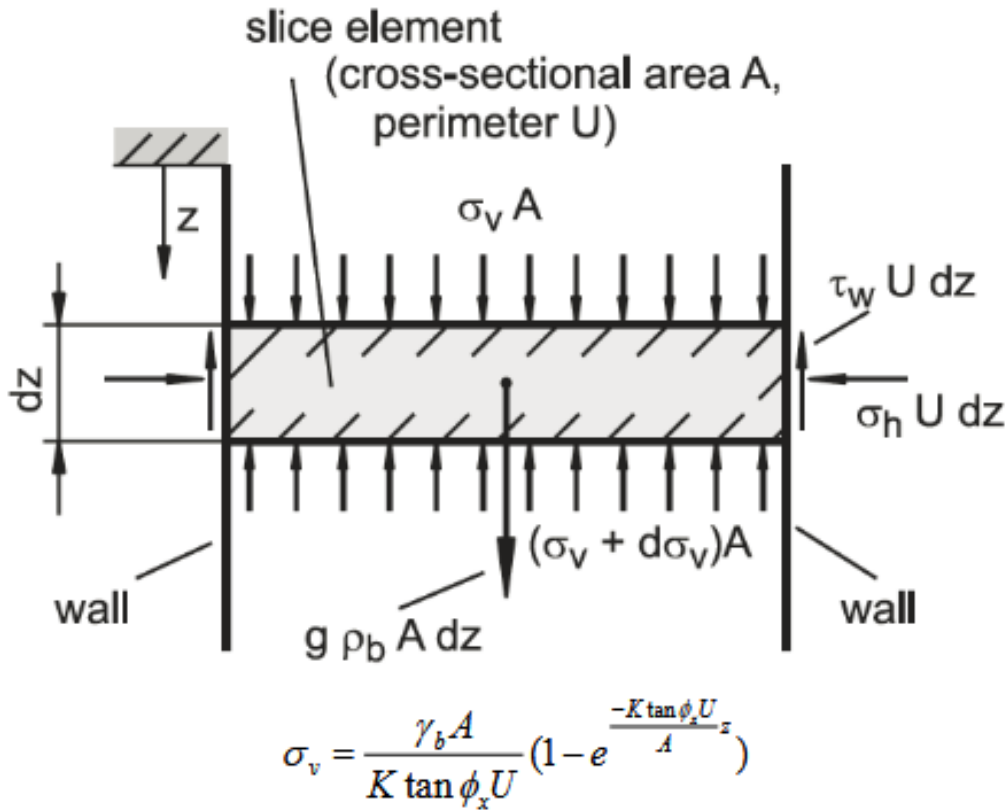
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Underground Mining and Bulk Materials Handling (MIN E – 630) Assignment #3

1 Question statement

In Lecture 4, a detailed process was demonstrated, using the slice element method to derive the formula for mine backfill stress under a “z-coordinate” system.

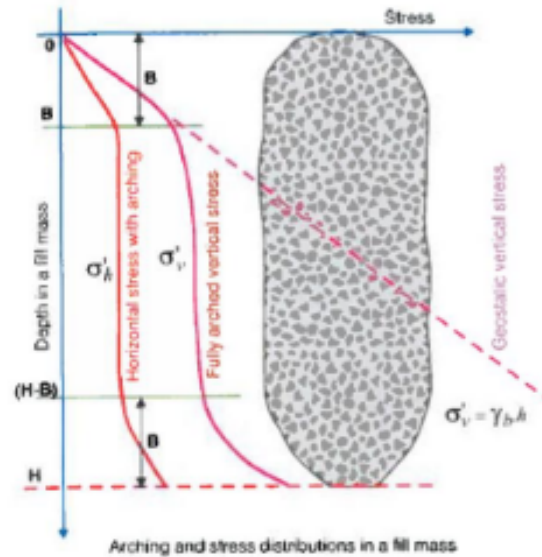
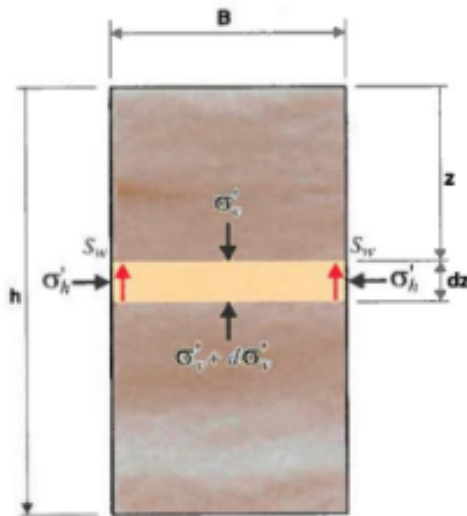


A = area of the stope (m^2); U = perimeter of the stope (m); z = depth below the surface (m);
 γ_b = Dry bulk unit weight of the bulk solid (N/m^3).
 σ_v = Vertical in-situ stress of the bulk solid (Pa)
 ϕ_x = Angle of wall friction ($^\circ$)

But in the current formula, the cohesion (C_x), Pa, between the mine backfill and the stope wall has been ignored.

1. As a graduate student, please help the mining industry derive an updated formula that includes the C_x . Please show your detailed process from the beginning to the end. (90 points)
2. The current industry is using the following formula. Using your critical thinking, please list the concerns. (10 points)

Arching in fill masses



$$\sigma'_v = \frac{(B\gamma_b - 4c')}{4K_0 \tan \phi'} \left(1 - e^{\frac{-4K_0 \tan \phi'}{B} z}\right)$$

B = width of the fill mass (m); c' = effective cohesion (kPa); z = depth below the surface of fill masses (m); γ_b = dry bulk unit weight of the fill mass (kN/m³).

1.1 1. Derivation process

Per Newton's second law, in this statement, assume material inside is static. so.

$$\sum F = 0$$

(EQ 1)

For this slice, expand ZQ1 along with z direction.

$$\sigma_v A + \gamma_b A dz = (\sigma_v + d\sigma_v) A + T_w U dz \quad \text{ZQ 2}$$

Where: $T_w = C_0 + \sigma_h \tan \phi$
 $\gamma_b = \gamma A$

Substitute T_w, γ_b and eliminate $\sigma_v A$.

$$\gamma_b A dz = A d\sigma_v + (C_0 + \sigma_h \tan \phi) U dz \quad \text{ZQ 3}$$

There is a relationship between σ_v and σ_h

$$K = \frac{\sigma_h}{\sigma_v}$$

ZQ 4

Substitute K into ZQ3 and re-arrange it

$$A d\sigma_v = \gamma_b A dz - C_0 U dz - K \sigma_v U dz \tan \phi \quad \text{ZQ 5}$$

Both sides divided by $A \cdot dz$

$$\frac{d\sigma_v}{dz} = \gamma_b - \frac{C_0 U}{A} - \frac{K U \tan \phi}{A} \sigma_v$$

let $m = \gamma_b - \frac{C_0 U}{A}$, $n = -\frac{K U \tan \phi}{A}$

$$= n \sigma_v + m$$

$$\Rightarrow \frac{d\sigma_v}{dz} - n \sigma_v = m$$

ZQ 6

For the first-order linear differential equation, the general form

$$\sigma_v = -\frac{m}{n} + C_1 e^{nz}$$

At the top of silo, $z=0$, $\sigma_v=0 \Rightarrow C_1 = \frac{m}{n}$

$$\text{Thus } \sigma_v = \frac{m}{n} (e^{nz} - 1)$$

ZQ 7

Substitute m, n, back into ZQ 7

$$\sigma_v = -\frac{\gamma_b A - C_0 U}{\frac{K U \tan \phi}{A}} \left(e^{-\frac{K U \tan \phi}{A} z} - 1 \right)$$

$$= -\frac{\gamma_b A - C_0 U}{K U \tan \phi} \left(e^{-\frac{K U \tan \phi}{A} z} - 1 \right) \quad \text{ZQ 8}$$

clear the negative sign

$$\sigma_v = \frac{\gamma_b A - c_0 U}{KU \tan \phi} \left(1 - e^{-\frac{KU \tan \phi}{A} z}\right) \quad \text{EQ9.}$$

Let $c_0 = 0$, we can get the same equation

$$\sigma_v = \frac{\gamma_b A}{KU \tan \phi} \left(1 - e^{-\frac{KU \tan \phi}{A} z}\right)$$

1.2 2. Comparison between two equations

The current industry is using the following formula.

$$\sigma'_v = \frac{(B\gamma_b - 4c')}{4K_0 \tan \phi'} \left(1 - e^{-\frac{4K_0 \tan \phi'}{B} z}\right)$$

Where:

- σ'_v : Width of the trench or silo (m).
- c' : Effective cohesion (kPa).
- z : Depth below the surface of fill masses (m).
- γ_b : Dry bulk unit weight of the fill mass (kN/m³).

Here is our EQ9.

$$\sigma_v = \frac{\gamma_b A - c_0 U}{KU \tan \phi} \left(1 - e^{-\frac{KU \tan \phi}{A} z}\right)$$

Compare them, we can find that: The current industry formula assumes that the silo shape is a square with a width of B. So the $\frac{A}{U} = \frac{B^2}{4B} = \frac{B}{4} \implies A = \frac{BU}{4}$, substitute into EQ9, divide the numerator and denominator by U, we can get $\sigma_v = \frac{B\gamma_b - 4c_0}{4K \tan \phi} \left(1 - e^{-\frac{4K \tan \phi}{B} z}\right)$

However, some key points require attention.

First, the current formula simplifies the silo shape. When we use the other shapes, like a circle or an ellipse, the extra error occurs.

Second, the effective cohesion is not totally from the materials or the wall of silo, it is determined by both of them at the same time.

Third, the γ_b might not be a constant; the materials at the bottom of silo might be slightly greater than the materials at the top, because the finer particles could fill into the fractures due to gravity.

Last, the process of removing material from the silo for use. The static friction between the material and the wall will become sliding friction. The formula might no longer be applicable.

These points require engineers to be concerned about them in real projects, dynamically and continuously.